



Front-end Development

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Web Technologies Basics

Concepts

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1. Web Page, Web Site and Web Application

Web Page

- Web page is a **document** or an **information resource** that are suitable for the World Wide Web
- It can be accessed through a **web browser** and displayed on a monitor or a mobile device
- This information is usually in **HTML** or **XHTML** format, and may provide navigation to other web pages via **hypertext links**

<div id="header">...</div>

- A web page frequently refer to **other resources**:
 - style sheets (CSS),
 - scripts (JavaScript),
 - and images

Web Site

- A web site is a **collection** of related web pages containing web resources
 - It have **common navigation** between web pages
 - It is **hosted** on at least one **web server**
 - It is accessible via a network
- **All publicly accessible websites** collectively constitute the **World Wide Web**

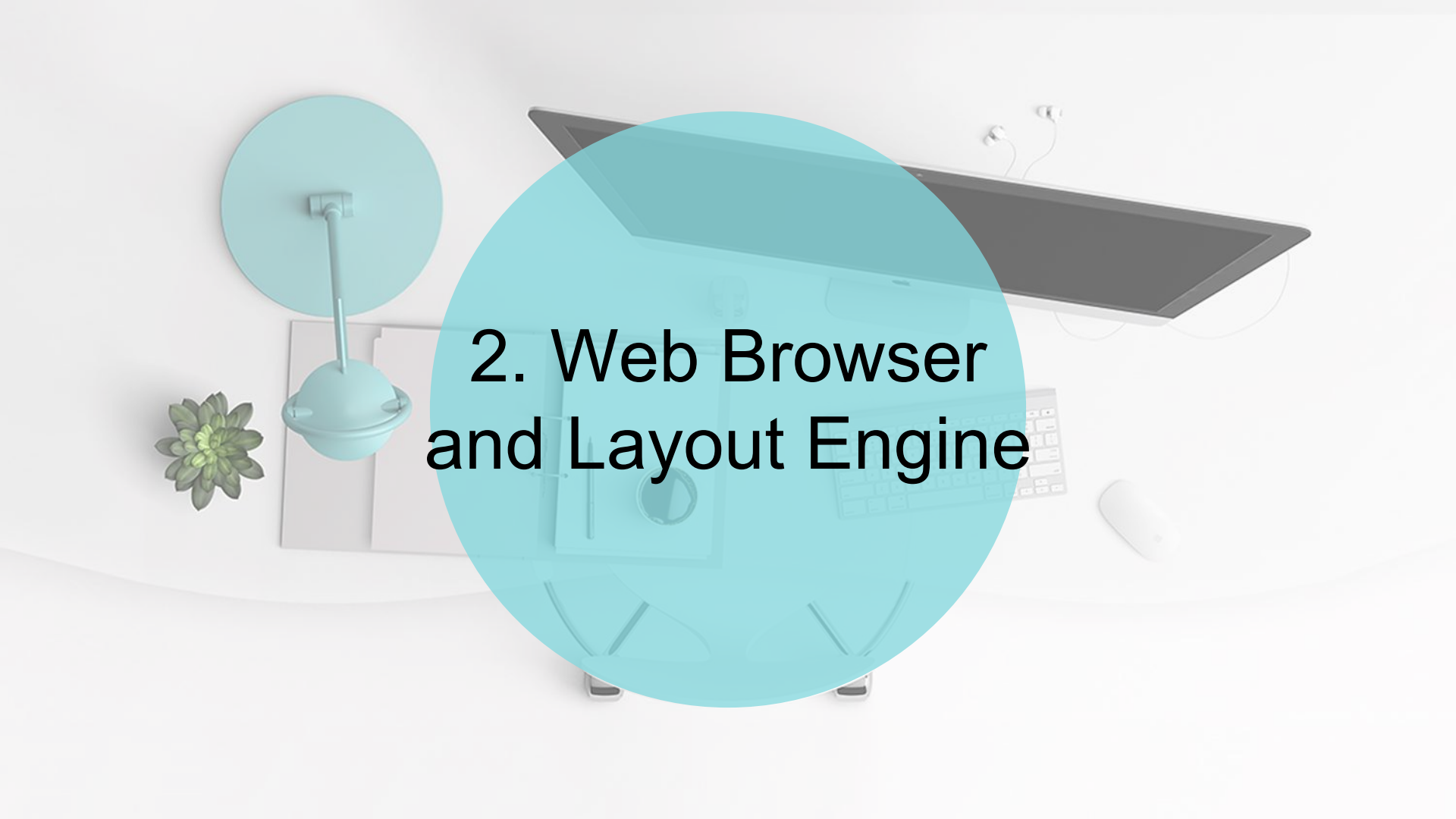
Web Application

- Next level web sites
- High interactivity
- High accessibility (Cloud)
- Desktop-like application in the web browser



Operating Principle





2. Web Browser and Layout Engine

Web Browser

- A Web browser is a program designed to enable users to **access, retrieve** and **view** documents and other resources from the Web
- Main responsibilities:
 - **Bring** information resources to the user
 - **Present** web content (render HTML, CSS, JS)
 - **Capable of executing** applications within the same context as the document on view (Flash)



Layout Engine

- A layout engine is a software component that **displays the formatted content** on the screen by combining:
 - Marked up content (HTML)
 - Formatting information (CSS)
- Typically embedded in web browsers, e-mail clients, online help systems or other applications that require the displaying (and editing) of web content
- The layout engine is the “**heart of a browser**”

Why are some elements displayed differently on different browsers?

chrome



EDGE



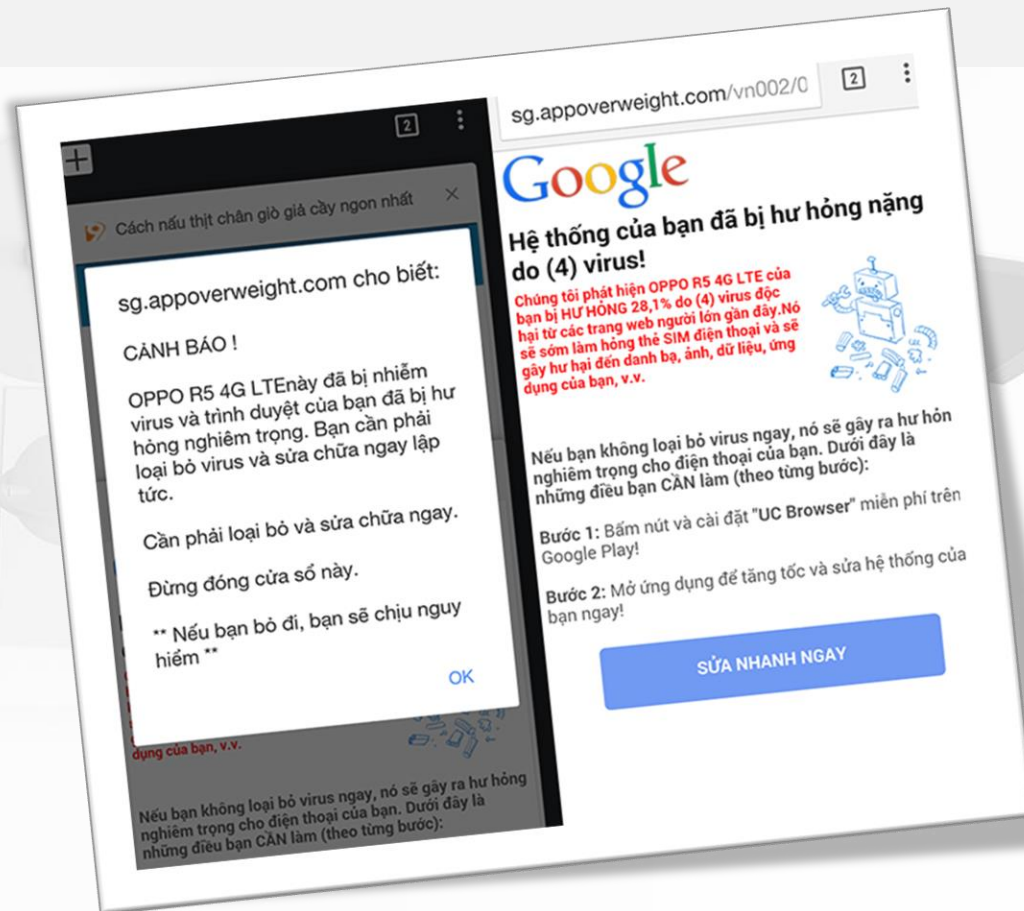
Firefox



Layout Engines and Web Browsers

- **Gecko**-based
Firefox, Netscape, SeaMonkey, etc.
- **Blink**-based
Chrome, Opera, Microsoft Edge (2018 and later)
- **WebKit**-based
Safari, iOS, Maxthon, Chrome (up to v27), etc.
- **Trident**-based
Internet Explorer, Netscape, Maxthon, etc.
- **EdgeHTML** (fork of Trident)
Microsoft Edge (2017)

How do websites know what device we are on?



User Agent String

- A user agent string **identify web browsers** and their version
- It can have some **additional information** like layout engine, user's operating system, etc.
- Example:

Mozilla/5.0 (**Windows NT 6.3; WOW64**) AppleWebKit/537.36 (KHTML, like Gecko)

Chrome/41.0.2272.118 Safari/537.36



3. Hardware Server

Hardware Server

- A hardware server is a **physical computer** dedicated to running one or more such services
- Servers are placed in colocation centers
- The server may be:
 - Database server
 - File server
 - Mail server
 - Print server
 - VPS server



Microsoft
IIS

NGINX

SERVER
COMBINATION

4. Web Server

Apache, IIS, nginx, lighttpd,...

APACHE

What do the Web Servers Do?

- A **web server** is a software product that uses the operating system to **handle web requests**.

Web server **serves Web content**.

- These requests are redirected to other software products (ASP.NET, PHP, etc.), depending on the web server settings



5. HTTP and HTTPS

Hypertext Transfer Protocol

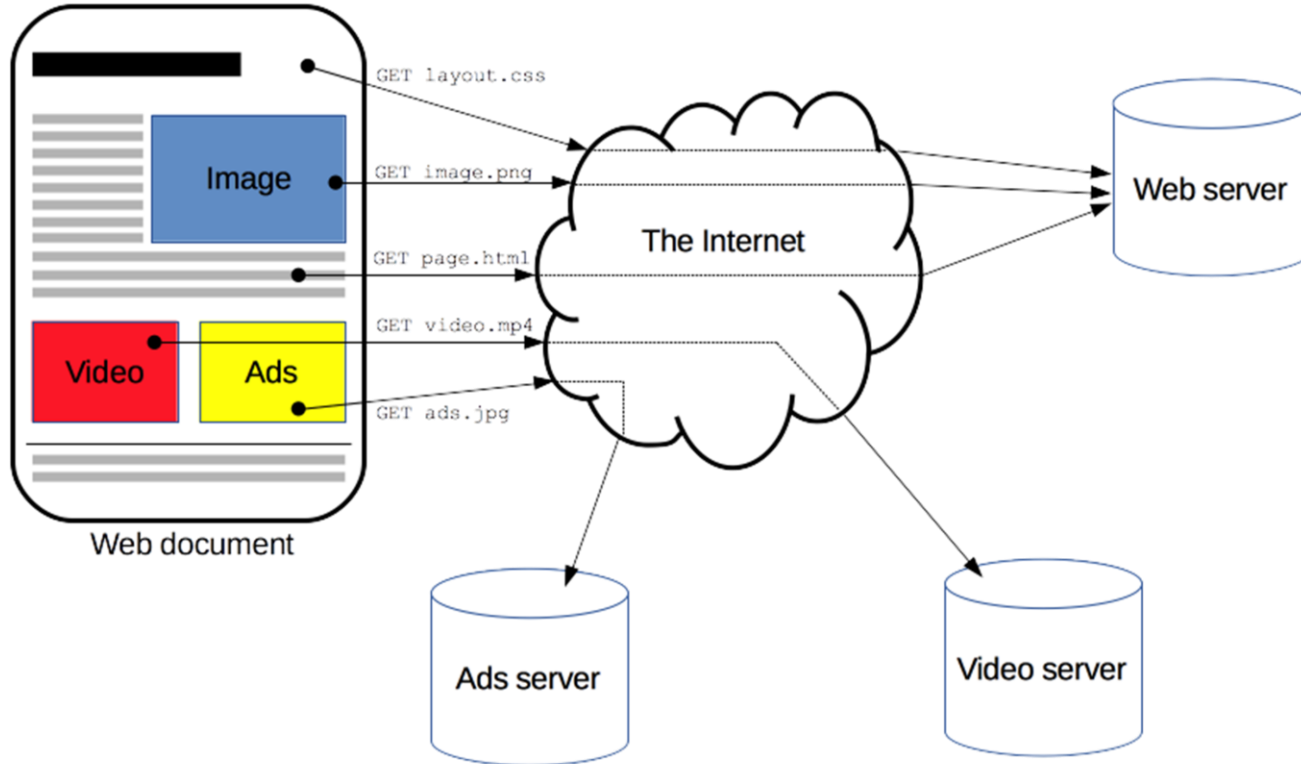
Hypertext Transfer Protocol

HTTP

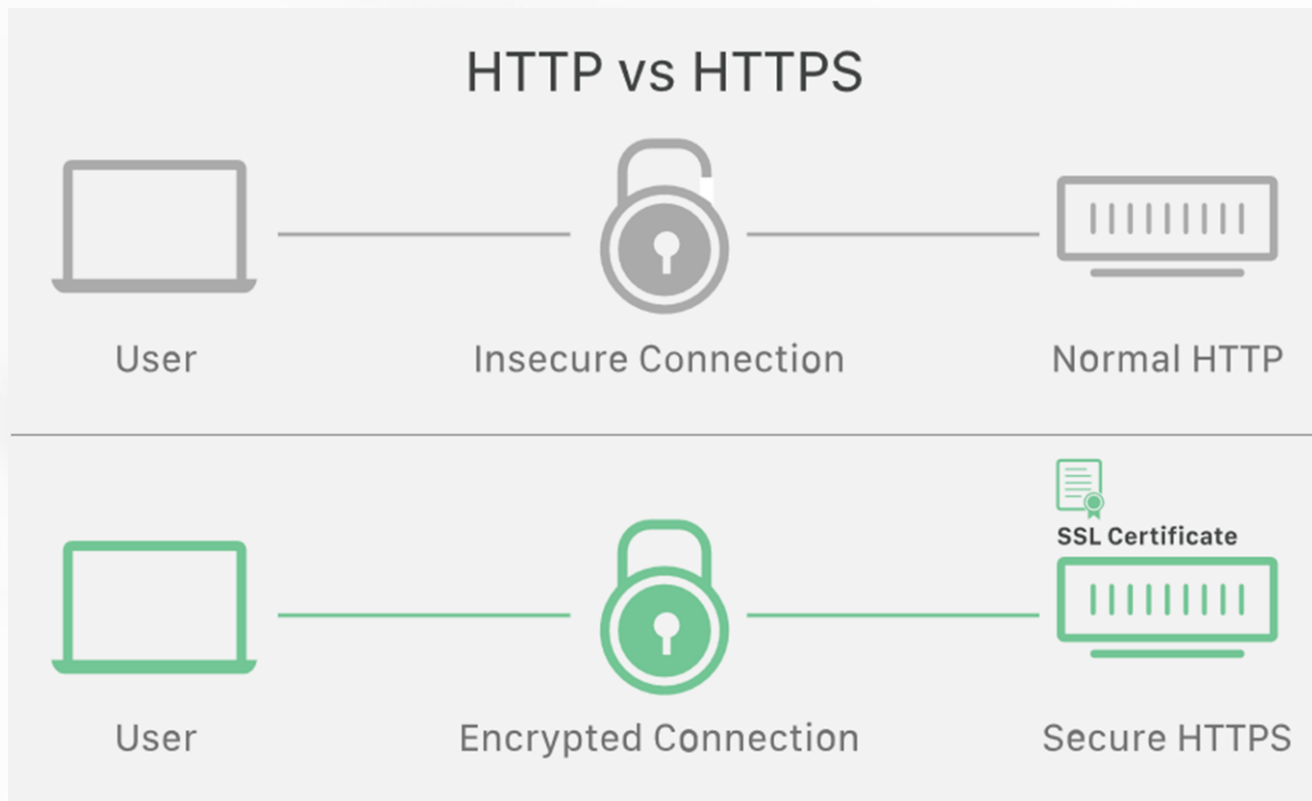
- HTTP is an application-layer **protocol** for **transmitting hypermedia documents**, such as HTML.
- It was designed for **communication** between web **browsers** and **web servers**.
- HTTP follows a **classical client-server model**, with a client opening a connection to make a request, then waiting until it receives a response.

Hypertext Transfer Protocol

HTTP

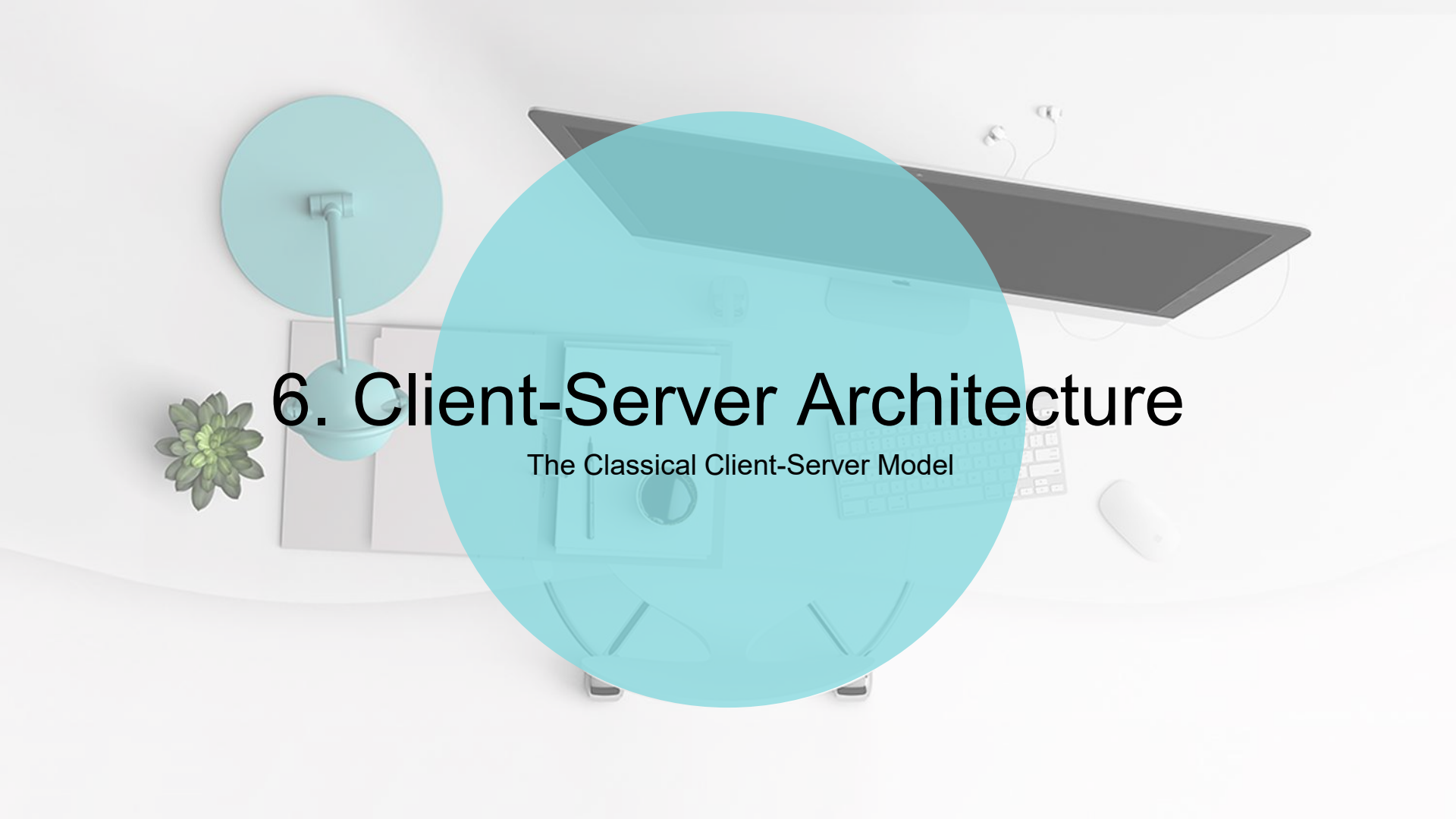


HTTPS



HTTPS

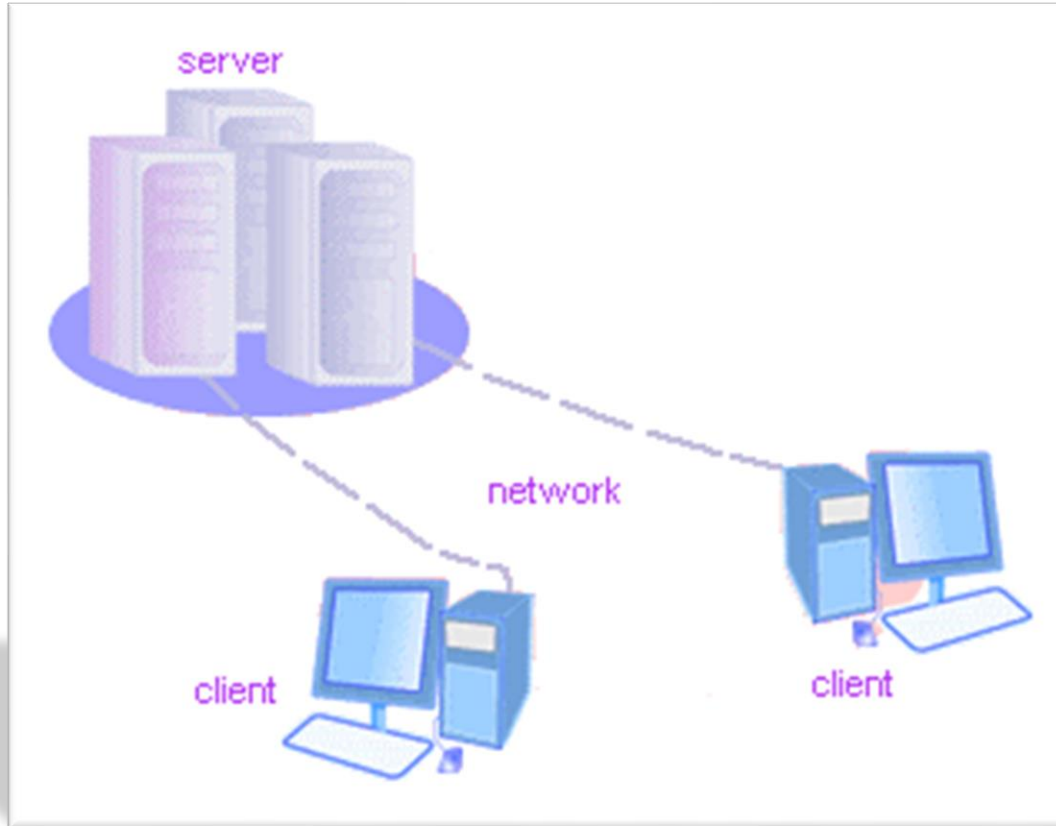
- HTTPS is HTTP **with encryption**.
- The only difference between the two protocols is that HTTPS **uses TLS (SSL) to encrypt** normal HTTP requests and responses.
- HTTPS is **far more secure** than HTTP.

A top-down view of a clean, white desk. On the left is a teal desk lamp with a circular base and a small green succulent. In the center is a large, semi-transparent teal circle. To the right of the circle is a wide, thin black monitor, a white keyboard, and a white mouse. A pair of white earbuds lies near the monitor. The background is a plain, light gray.

6. Client-Server Architecture

The Classical Client-Server Model

Client-Server Architecture



Client-Server Architecture

The client-server model consists of:

- A **server**

- A single machine or cluster of machines that provides web applications (or services) to multiple clients
- Examples:
 - ✓ Web server running PHP scripts or ASP.NET pages
 - ✓ IIS-based Web server
 - ✓ WCF-based service
 - ✓ Services in the cloud

Client-Server Architecture

The client-server model consists of:

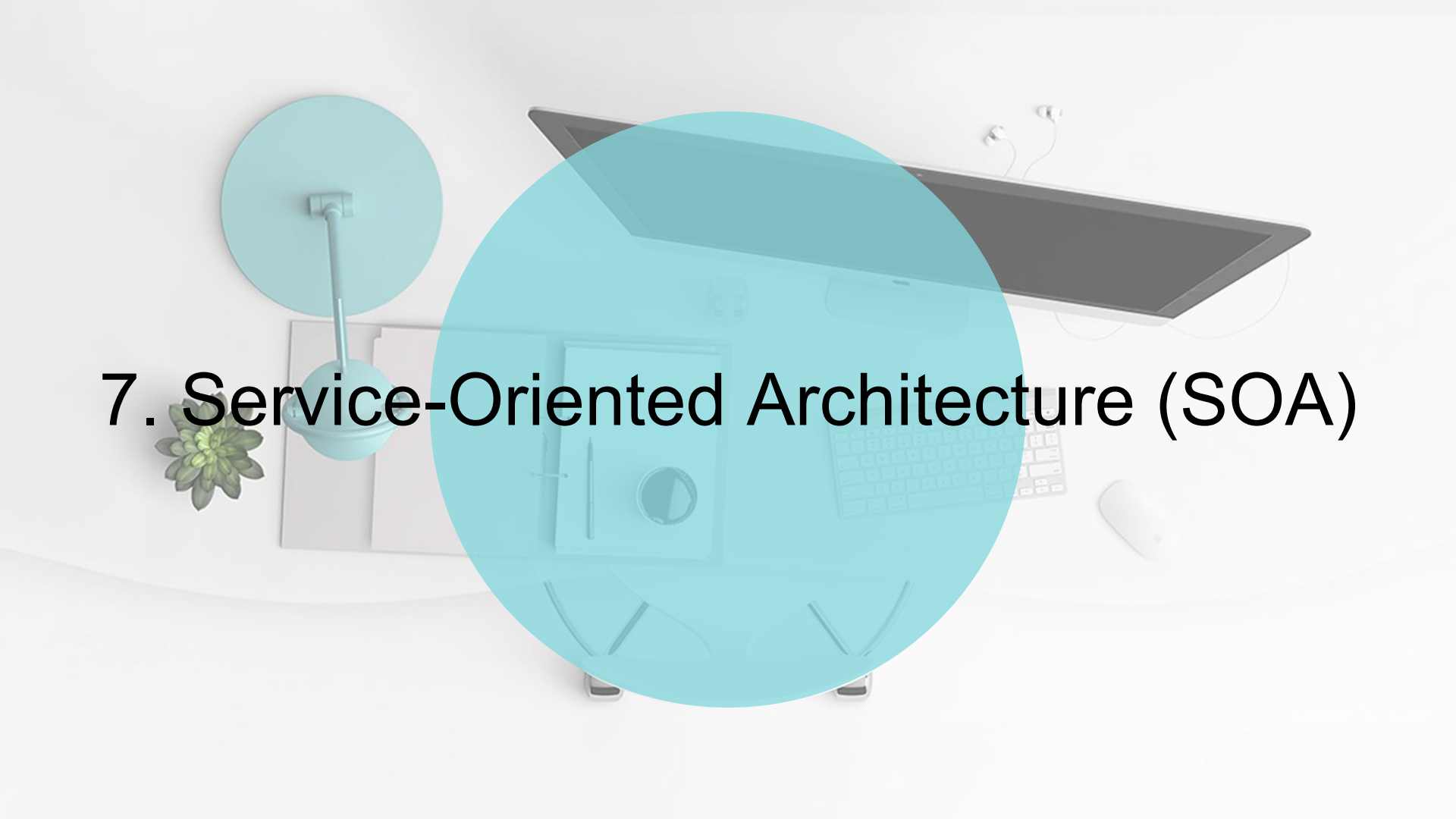
- A **client**

- A software application that provides UI (front-end) to access the services at the server
- Examples:
 - ✓ Web browsers
 - ✓ Desktop applications

Client-Server Model

Examples

Server	Client
Web server (nginx, Apache, IIS)	Web browser
FTP server (ftpd, File Zilla Server)	FTP client (FileZilla Client)
EMail server (Microsoft Exchange Server)	Email client (Outlook)
SQL Server	SQL Server Management Studio
DNS server (bind)	DNS client (resolver)

A top-down view of a clean, white desk. On the left is a teal desk lamp with a circular base and a small green succulent. In the center is a large, semi-transparent teal circle. To the right of the circle is a large black monitor, a white keyboard, and a white mouse. A small white container with a pen and a circular object sits on the desk. The background is a plain white surface.

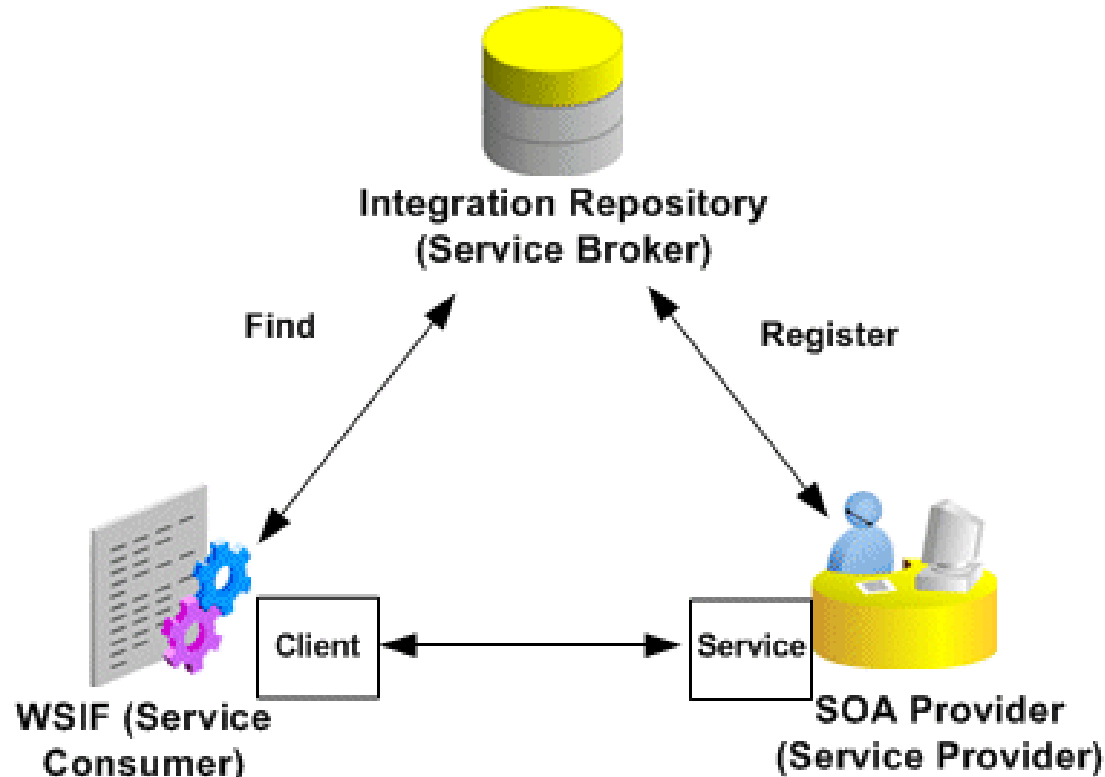
7. Service-Oriented Architecture (SOA)

What is a Service?

In the real world, a “service” is:

- A piece of work performed by a service provider
- Provides the client (consumer) some desired result by some input parameters
- Easy to use
- Always available
- Has quality characteristics (price, execution time, constraints,...)

Service-Oriented Architecture



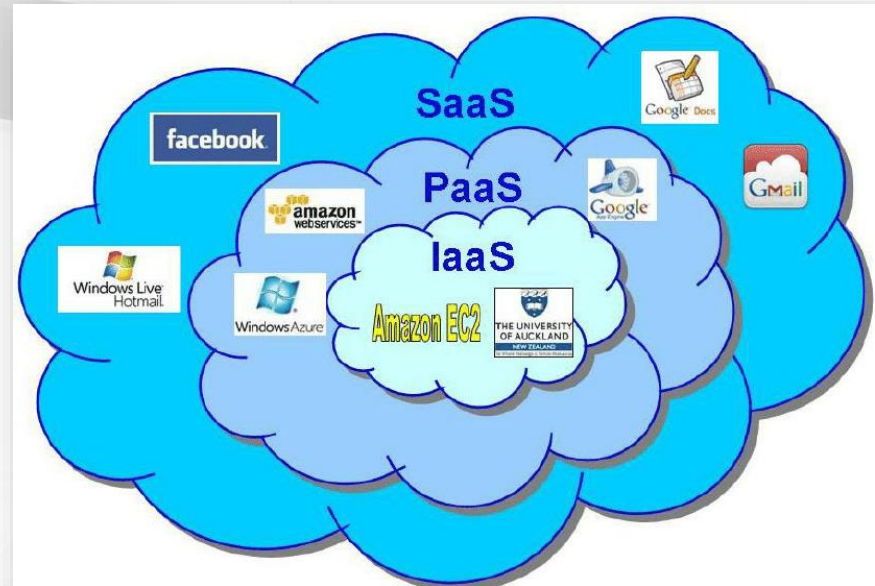
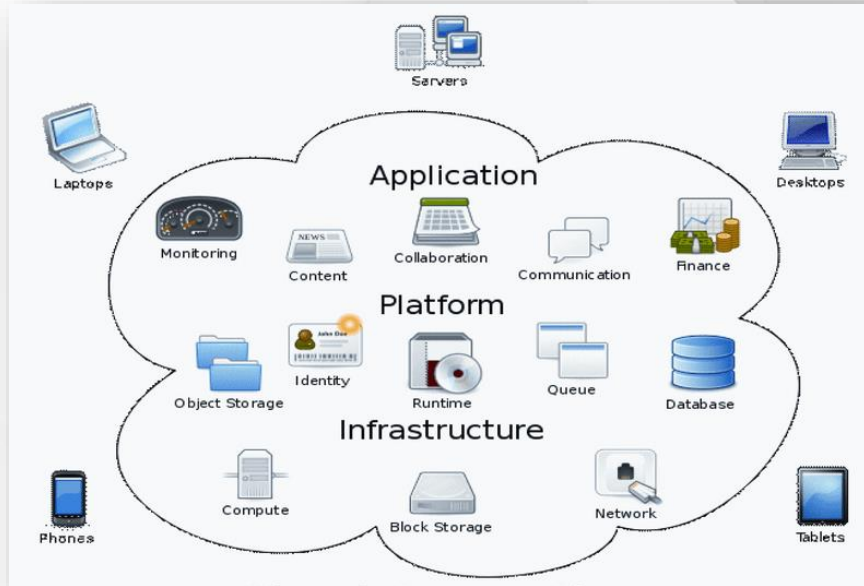
A top-down view of a clean, white desk. On the left is a teal desk lamp with a circular backplate and a small green succulent. In the center is a large, semi-transparent teal circle containing the text. To the right of the circle is a wide, thin black monitor, a white keyboard, and a white mouse. A small white cube sits on the desk near the monitor. The background is a plain white surface.

8. What is “Cloud”?

What is Cloud?

- **Cloud** $\approx$ multiple hardware machines combine their computing power and resources
 - **Share** them between multiple applications
 - To **save costs** and **use resources** more **efficiently**
- **Public clouds**
 - Provide computing resources on demand
 - ✓ Publicly in Internet
 - ✓ Paid or free of charge (to some limit)
 - Amazon AWS, Google App Engine, Microsoft Azure, Rackspace, PHPFog, Heroku, AppHarbor

Cloud Computing Models



Cloud Computing Models

▪ Infrastructure as a Service (IaaS)

- Virtual machines in the cloud on demand
- Users install the OS and software they need

▪ Platform as a Service (PaaS)

- Platform, services and APIs for developers
- E.g., Java + JBoss + JSF + JPA + MongoDB or JavaScript + Node.js + MongoDB + RabbitMQ

▪ Software as a Service (SaaS)

Hosted application on demand (e.g., WordPress)



Q&A